

# Lab 1 - Setting up your laptop for Artificial Intelligence and Machine Learning (AIML)

Set up and install the software used for AIML. We will be using Anaconda software to create python environment and install 3rd party libraries/packages such as numpy, scikit-learn, keras, TensorFlow and etc. Some of these libraries/packages update frequently and version differences might cause working Python program to stop working.

Follow all the instructions to do the installation. After successful installation, use the python test codes found in the last page of this laboratory sheet to ensure you can run and get the expected output.

## Install Anaconda if you have not installed before

Go to <https://docs.anaconda.com/anaconda/install> to download the version for your OS. Follow the installation instructions to install the program. Accept all default settings.

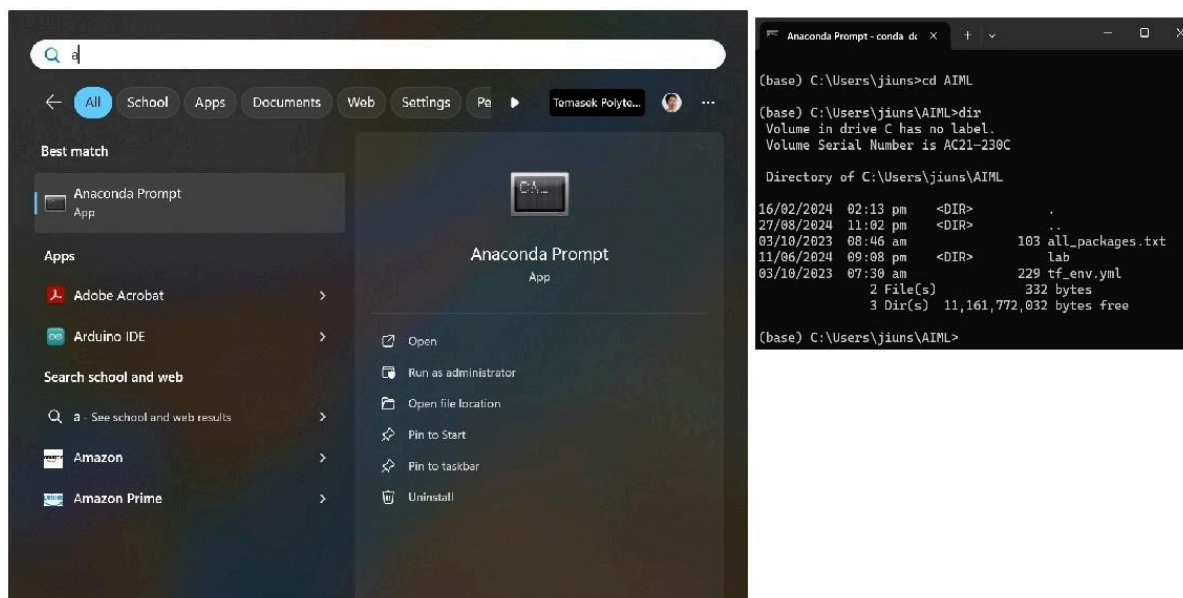
## Create environment

Create a "AIML" folder in the root directory. "C:\Users\name".

Download **tf\_env.yml** file that can be found under the Installation & Setup folder in TP LMS AIML website. Save the file to the folder you have just created. Example: "C:\Users\name\AIML"

Open anaconda prompt and change directory to "C:\Users\name\AIML" using the following command

## cd AIML



Use the command **dir** to check that you have downloaded the **tf\_env.yml** file in this directory.

Run the following command to create the TensorFlow environment in Anaconda with the name **tf**, where we will install all the required 3rd party libraries/packages.

## conda env create -f tf\_env.yml

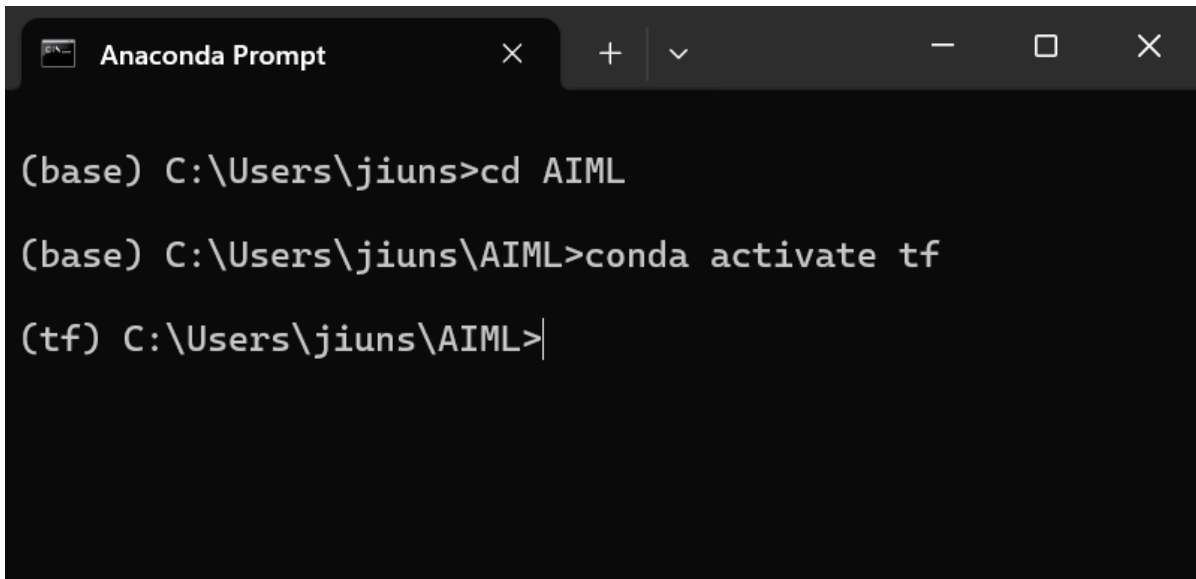
Below are all the libraries/packages installed in the environment.

- jupyterlab
- matplotlib
- numpy
- scikit-learn
- pandas
- seaborn
- keras
- tensorflow
- flask
- Werkzeug

### Activate the environment

After successful installation, every time you open an anaconda prompt, you need to activate the **tf** environment, if not the **base** default environment will be used.

**conda activate tf**



```
Anaconda Prompt
(base) C:\Users\jiuns>cd AIML
(base) C:\Users\jiuns\AIML>conda activate tf
(tf) C:\Users\jiuns\AIML>
```

To add additional package to your environment. You need to open an anaconda prompt and activate the environment and run the install package command.

The following demonstrates how to install TensorFlow\_datasets package into the **tf** environment.

**(base) C:\Users\name\AIML>conda activate tf**

**(tf) C:\Users\name\AIML>pip install tensorflow\_datasets**

### Launch Jupyter Notebook

Open anaconda prompt.

Change the directory to AIML.

**(base) C:\Users\name>cd AIML**

Activate the environment that you have created previously by running the following command.

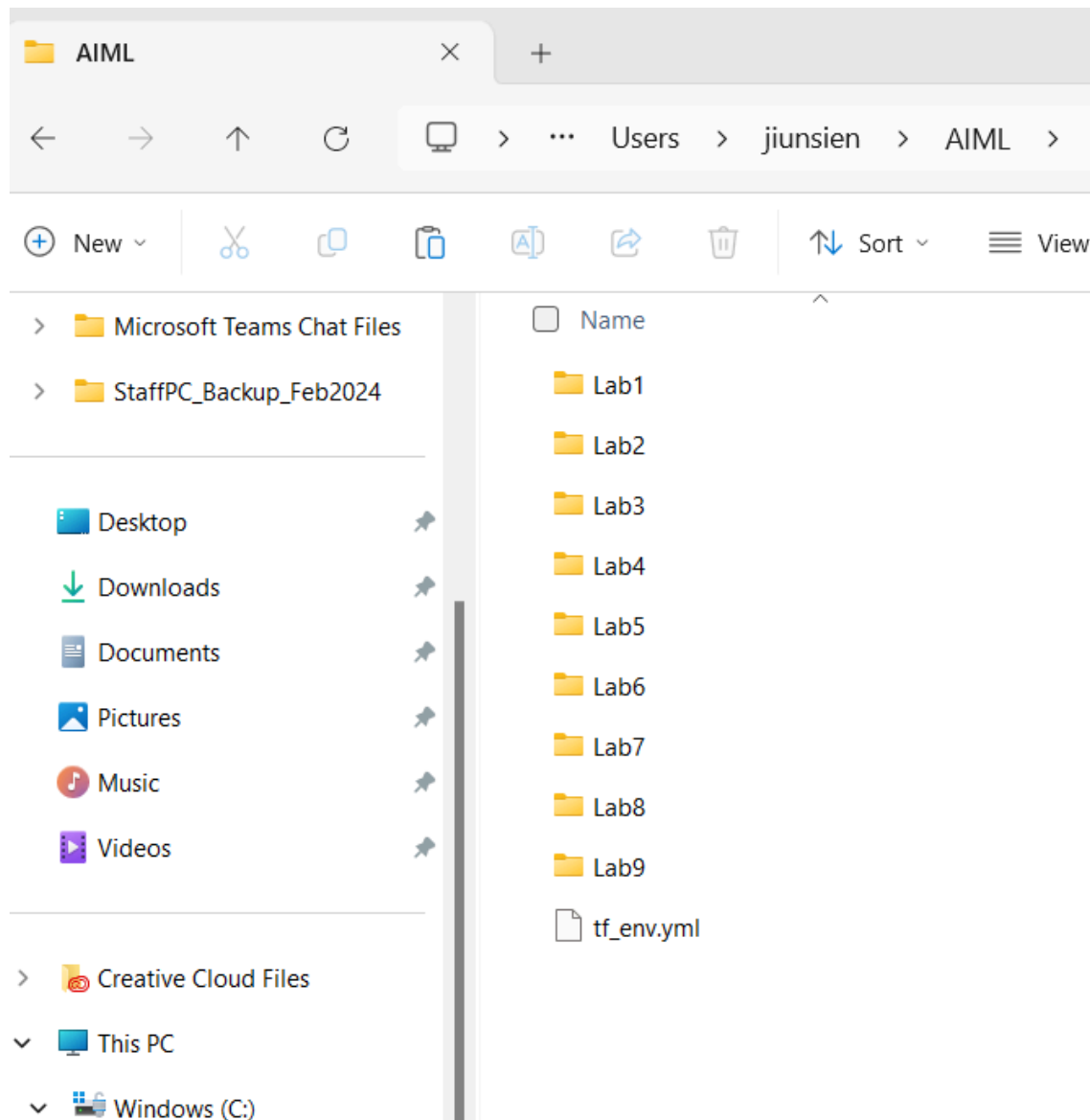
**(base) C:\Users\name\AIML>conda activate tf**

Protecting jupyter notebook environment with authentication. Type the following command to set password for the jupyter notebook.

**(tf) C:\Users\name\AIML>jupyter notebook password**

*Note: While keying the password, the cursor at the command prompt will not move until you hit the **Enter** button when you have finished keying your password. DO REMEMBER YOUR PASSWORD.*

For good practices, create and save all your laboratory folders (e.g. Lab1, Lab2, Lab3, etc.) under the "C:\Users\name\AIML" directory from this week onward.



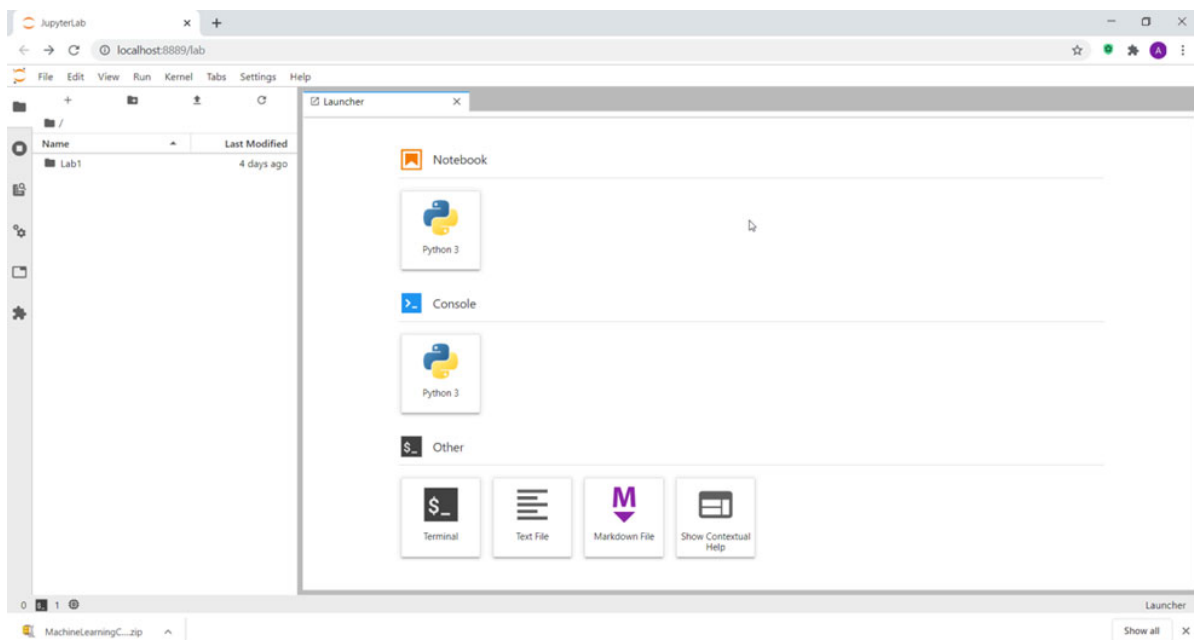
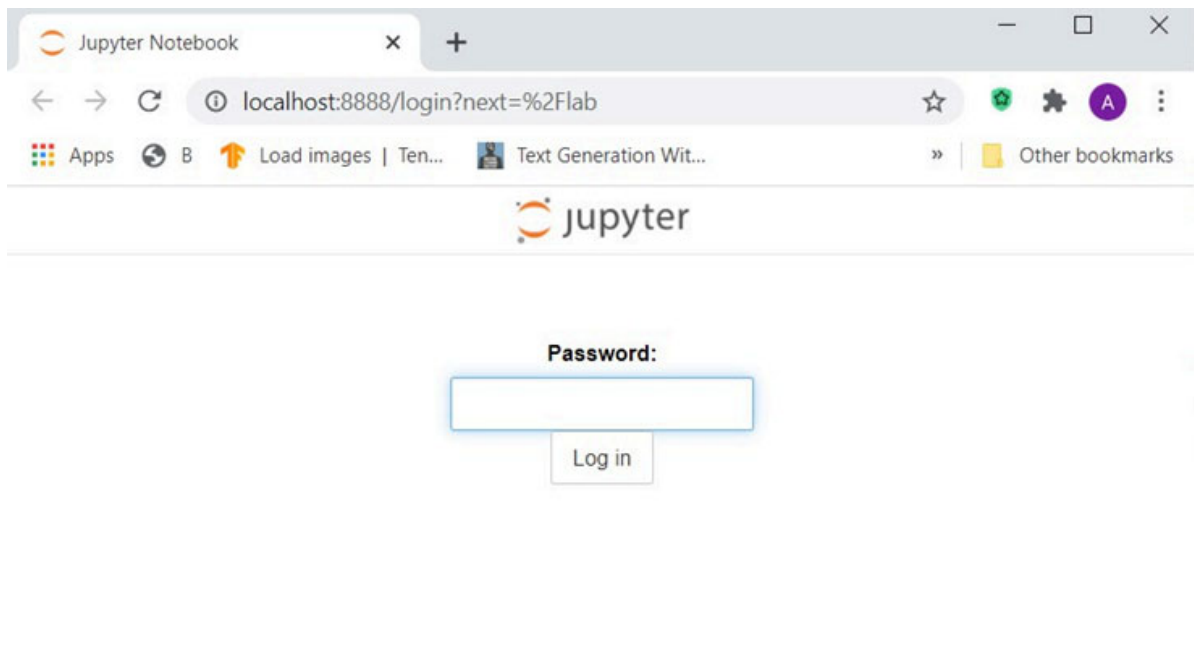
After finished creating the lab folders, you may download all the softcopy of the laboratory sheets from the laboratory folder in TP LMS AIML website.

We will be using jupyter notebook as our laboartory sheet for this subject. Jupyter lab is an interface to view the notebooks in the directories.

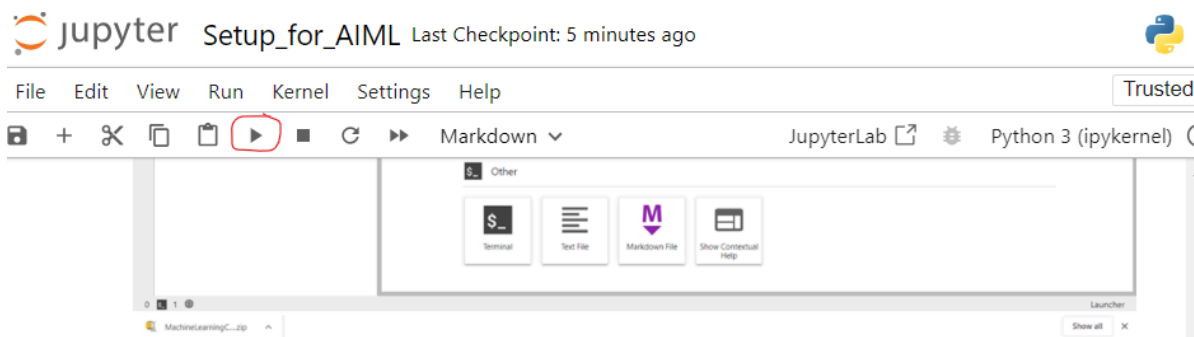
To launch the jupyter lab, run the following command in anaconda prompt after you change your directory and activate the **tf** environment.

**(tf) C:\Users\name\AIML>jupyter lab**

Type the password that you have created to log into the jupyter lab environment.



Click the play icon to run the below cell by cell.



```
In [8]: # The following code is to suppress the warnings on code that are deprecated in future version of py
import warnings
warnings.simplefilter(action='ignore', category=FutureWarning)
```

```
In [9]: # The following code is to check the version of TensorFlow you have installed.
import tensorflow as tf
print(tf.__version__)
```

2.16.1

```
In [10]: # The following codes check whether keras.utils and sklearn Libraries are working properly.
from sklearn import preprocessing
from tensorflow.keras.utils import to_categorical
#Label + Onehot Encoder
Y = ['unacc', 'acc', 'acc', 'vgood', 'unacc', 'acc', 'unacc', 'vgood', 'unacc', 'acc', 'good', 'vgood']
encoder = preprocessing.LabelEncoder()
encoder.fit(Y)
#print(list(encoder.classes_))

encoded_Y = encoder.transform(Y)
dummy_y = to_categorical(encoded_Y)
print(dummy_y)
```

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[[0. 0. 1. 0.]
 [1. 0. 0. 0.]
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In [ ]:
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